Goal: predict risk for an individual patient as new biomarker measurements arrive.

challenges: 1. errors in marker measurements
2. the intermittent visits.

Cox model needs each marker's true value at each event time.

↳ but what is Cox? (semi-parametric)

a regression model to predict time-to-event data.

a model with various covariates (X) (e.g. risk factors) affecting survival time.

- it ignores the exact shape of the baseline hazard over time. (mathematically cancels out)
- constant risk ratios over time.

explanatory variables ↘ ↙ coefficients

$$h(t \mid X) = h_0(t)\, \exp(X\beta)$$

↑ hazard func.

$$\text{Hazard Ratio} = \exp(\beta_j)$$

↑ baseline hazard
→ >1 increase the hazard
<1 decrease the hazard

To find the best-fit coefficients, Partial likelihood Estimation and an optimization loop is used.

↳ Newton-Raphson Algorithm

$$LL(\beta) = \sum_{i=1}^{K} \left[X_i\beta - \ln\left(\sum_{j \in R_i} e^{X_j\beta} \right) \right]$$

↙ log partial likelihood

↘ Risk set (active/alive patients)

Score func. (first derivative) ←

$$U(\beta) = \sum_{i=1}^{K} \left[X_i - \frac{\sum_{j \in R_i} X_j\, e^{X_j\beta}}{\sum_{j \in R_i} e^{X_j\beta}} \right]$$

information matrix (second derivative) ←

$$I(\beta) = \sum_{i=1}^{K} \left[\frac{\sum_{j \in R_i} X_j^2\, e^{X_j\beta}}{\sum_{j \in R_i} e^{X_j\beta}} - \left(\frac{\sum_{j \in R_i} X_j\, e^{X_j\beta}}{\sum_{j \in R_i} e^{X_j\beta}} \right)^2 \right]$$

iteration update

$$\beta_{new} = \beta_{old} + \frac{U(\beta_{old})}{I(\beta_{old})}$$

repeat until $U(\beta) \approx 0$

---

baseline hazard ← $h_0(t_i) = \dfrac{d_i}{\sum_{j \in R_i} e^{x_j \beta}}$

d_i → # of events occurred at time t_i.

↳ total sum of risk weight.

↳ Breslow Estimator

---

when we don't have all values for markers at all time, Joint models fix it.

challenge: many markers (more than 10) makes the model computationally intractable.

Question To Solve?

How to get full accuracy of a full joint model at a cost that scales to many markers?

Concept 1: ↳ linear mixed model (longitudinal submodel) (baseline)

handles how a patient's health markers change over time.

for marker k, patient i, time s:

$$Y_{ijk} = \underbrace{\eta_{ik}(s_{ijk} \mid \beta_k, b_{ik})}_{= \eta_{ik}(s) + \varepsilon} + \varepsilon_{ijk}$$

↙ measurement error

s_{ijk}: time

↘ random effects

↑ fixed effects

$i = 1, \ldots, N$
$j = 1, \ldots, n_i$
$k = 1, \ldots, K$

$$\eta_{ik}(s \mid \beta_k, b_{ik}) = x_{ik}(s)^T \beta_k + z_{ik}(s)^T b_{ik}$$

← η_{ik} : true, error-free value for patient's marker at time s.

Fixed effect: constant parameter for entire population. doesn't change person to person.

Random effect: how much an individual deviates from population.

Concept 2: Time-to-event sub-model (given a patient's history, current trajectory, what's risk at time t?)

$$\partial_i(t \mid w_i, b_i) = \partial_0(t) \exp\left\{ \gamma^T w_i + \sum_{k=1}^{K} \alpha_k \underbrace{g_k(b_{ik}, x_{ik}, z_{ik}, \alpha_k, t)}_{\eta_{ik}(t)} \right\}$$

↑ hazard ↑ baseline hazard ↑ standard patient traits (e.g. age, sex) ↑ association parameter ↑ current true value of marker at time t dynamic evolving biomarker trajectories.

it is just formal way of showing that survival model's input is constructed directly out of longitudinal model's components.

Concept 3 : The standard two-stage approach and its two flaws.

① Step 1 by longitudinal submodel, patient's predicted health path over time is calculated.

Step 2 : takes score, plugs it in, spits out death risk.

[lab-data] → 1st model → [health score] → 2nd model → [death risk]

↙

ran by itself, it had no idea some sick patients died & dropped out of the study.

hence gives the 2nd model an optimistic health score, leading to biased result.

So the fix in the Paper's model is here in the first step.

② Stage 2 might be a victim again for stage 1 ignoring uncertainty.

Paper's method : Two-stage Joint model.

→ instead of K mixed models, fit K separate one-marker joint models. estimated with Bayesian MCMC in **JAGS**

→ Let the survival data "Talk" to the trajectory.

even if some data points are missing because the patient dropped out, the joint model uses the survival timeline to adjust & correct the trajectory line.

Bayesian .

in standard statistics parameters like a patient's true trajectory or regression coefficients are treated as fixed, constant, single unknown that we try to guess.

in Bayesian statistics, parameters are treated as random variables with probabilities.

Posterior Distribution : probability distribution of an unknown parameter after you have looked at your data.

$$P(\text{Parameters} \mid \text{Data})$$

↳ given lab visits & survival time
what is the full range of probabilities for what this true health trajectory parameters (b_i) actually are.

instead of a rigid number ⇒ a curve, probabilities.

Bayes' Theorem applied in joint models :

$$P(\theta, b_i \mid \text{Data}) = \frac{P(\text{Data} \mid \theta, b_i)\, P(\theta, b_i)}{P(\text{Data})}$$

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

The prior : what we believe about the parameters before looking at the patient's data.

$P(\theta, b_i)$

The likelihood : longitudinal & survival data are multiplied together.

$P(\text{Data} \mid \theta, b_i)$

$= P(\text{Longitudinal Lab Scores} \mid b_i) \times P(\text{Survival Time } T_i, \delta_i \mid b_i) = \partial(T_i \mid b_i) \times S_i(T_i \mid b_i)$

↑ how well the trajectory fits lab visits. ↑ how well hazard model predicts death

The posterior : updated probability distribution after prior & likelihood have merged.

$P(\theta, b_i \mid \text{Data})$

↳ a multi-part score measuring how well a guessed trajectory fits both the lab visits and the survival outcome.

Mechanism of how death corrects trajectory?

Test : computer tests a guess for a patient who died at year 1.5 but guesses a slow "healthy" slope.

slope of +0.2

round 1: 1. Does early lab visits fit 0.2 slope? sort of. (Score = 0.8)

2. Does +0.2 explain why one dies at year 1.5 ? NO. (Score = 0)

$0.8 \times 0 = 0$ ← likelihood.

updates guess to +1.2

round 2: 1. Score 0.4

2. Score 0.9 ← a steep slope meaning crashing fast.

likelihood = $0.4 \times 0.9 = 0.36$

Prior = 0.2

Posterior = $0.36 \times 0.2 =$ 0.072. > 0

bad guess thrown out. Keeping high posterior scores.

then surviving guesses are averaged.

So the posterior is a probability distribution. with mean & SD.

one might naively believe that the mean, only a number would be given & fed into the Cox model. however, by using ...

Multiple imputation happens.

Rubin's Rule.

→ M=100. 100 different plausible health scores are pulled out. (from the posterior distribution, we pull single concrete numerical values from the prob. curve.)
Cox model is run 100 times. (separately)

Rubin's rule works as a mathematically rigorous way to pool the dataset together. (we get 100 different output from Cox models. 100 various α_i $i = 1, ..., 100$)

(we also have variances (internal standard error))

Rubin's Recipe so 100 drawings.

Then refitting the Cox model M times & pool variances using Rubin's Rules.

finite imputation correction factor

total variance $T = \bar{W} + (1 + \frac{1}{M}) B$ ← between imputation variance.

↳ within imputation variance

used to calculate total honest variance & standard error.
if M large → buffer shrinks
if M small → buffer is large

↳ # of imputations
$M \to \infty$: 1+0 = 1.0
M = 5 : 1 + 1/5 = 1.2
M = 100 : 1 + 1/100 = 1.01

T : average of the variances calculated inside each Cox model run.

M : Total # of datasets created by drawings. (M = 100)

B : how much estimated parameter bounced around across M runs.

$1 + \frac{1}{M}$: since # of imputations is finite, this factor corrects for any extra random noise introduced by stopping at m. mathematical buffer

$$B = \frac{1}{M-1} \sum_{m=1}^{M} (\hat{\alpha_m} - \bar{\alpha})^2$$

↳ if α's are similar (1.41, 1.42, 1.4) B is close to zero (certain)
if α's are very various (1.1, 1.85, 1.25) B is high. (high uncertainty)

Rubin does these: acts like a blender

1. averages α's $\bar{\alpha} = \frac{\alpha_1 + \alpha_2 + ... + \alpha_{100}}{100}$ let's say $\bar{\alpha} = 1.48$
2. takes the variances & combine them using 2 uncertainties. of $\bar{W}$, B

   So mainly Rubin is honest about the error when we average α's. Since we take them from a probability dist.
3. $\bar{W}$ is the average of all variances that come with the 100 draws. ($s_1^2, s_2^2, ..., s_{100}^2$)

$$\bar{W} = \frac{1}{M} \sum_{m=1}^{M} s_m^2$$

So basically, probability distribution gives us guesses that are educated estimates, not absolute facts. they inherently carry uncertainty (doubt)

Rubin's Rule takes that doubt, measure it across multiple runs, & hand you an honest error margin alongside final average.

Uncertainty is the width of our doubt!

probability is the language we use to describe uncertainty.

accounting for uncertainty (error propagation) means:

" Not letting upstream models lie to downstream models. "

So in the end we take square root of T to have the honest Standard error.

$$SE = \sqrt{T}$$

$$\text{then} \quad Z = \frac{\bar{\alpha}}{SE}$$

standard critical value for 95% confidence interval ↙

and the final confidence interval $CI = \bar{\alpha} \pm (1.96 \times SE)$

Dynamic Prediction : risk score that updates in real-time as a patient comes back for new clinic visits & takes new lab tests.

(conditional probability)

given that the patient has successfully survived up to landmark time s,

and given all their longitudinal lab data up to that point Y(s)

what is the probability that they will survive up to a future time t?

$$\pi(t \mid s) = P(T > t \mid T > s, Y(s))$$

↑ future horizon — how far ahead we want to predict.

↑ landmark time — how long patient has survived.

↑ true event time.

↑ all the bumpy lab measurements collected up to time s.

$$= \exp(-H)$$

Cox model (Standard survival models) can only calculate a patient's risk at the very beginning of the study. (t=0) they cannot ingest new lab results over time.

So we have population parameters, now we want individual Patients' prediction.

---

fundamental law of survival analysis?

Connecting accumulated danger (cumulative hazard) to Survival probability.

in probability theory (Poisson process) if an event happens at an expected total rate H, the probability of experiencing zero events (zero death events means survival)

$$\frac{S(s+t)}{S(s)} = \frac{\exp(-H(s+t))}{\exp(-H(s))}$$

$$= \exp(-(H(s+t) - H(s)))$$

$$= \exp(-\Delta H)$$

↳ Risk = 1 − chance of survival

$$P(\text{Events} = 0) = \frac{e^{-H} \cdot H^0}{0!} = e^{-H}$$

$$S(t) = e^{-H(t)}$$

exp: between year 2 & year 4 : $\Delta H = H(Y4) - H(Y2) = 0.448$

$$\pi_i(4|2) = \exp(-\Delta H) = e^{-0.448} = \mathbf{63.9\%}$$

(stage 1) Posterior distribution : biomarker trajectory (what's the hidden biomarker slope in computer)

(stage 2) Hazard Rate (Cox model) : The rate of risk $\lambda(t)$ (at what speed is danger striking)

Survival Probability (dynamic prediction) : Patient's life/death outcome $S(t)$ (what's the chance of patient still breathing in 2 years)

Algorithm 1 is defined above. called first order.

means calculating the average slope ($\hat{b}_i$)

Plugs in mean posterior directly into the survival formula.

Algorithm 2 (Monte Carlo) (rigorous)

instead of plugging in the averages, runs a simulation loop to give you both a risk score & an honest prediction interval. (error bars)

instead of Alg. 1 taking average slope ($\hat{b}_i$) & association parameter (α) as 100% frozen known facts, it calculates uncertainty bounds.

Monte Carlo .

Loops L times (e.g. 500 draws) sampling from parameter uncertainty. ($N(\theta, T)$) and random effect distributions. ($b_{ik}^{(l)} \sim p(b_{ik} \mid T_i > s, Y_{ik}(s))$)

example round 1 : $\alpha^{(1)} = 0.46$ $\theta^{(1)} \sim N(\theta, T)$

$m_i^{(1)} = 2.8$

$$\text{Risk}^{(1)} = 1 - \frac{S^{(1)}(s+t)}{S^{(1)}(s)} = 1 - \exp(-\Delta H^{(1)}) = 30.4\%$$

there are 500 outputs & we take averages for final risk.

Computer then sorts 500 results ascending orders & gets 2.5th & 97.5 percentile for 95% prediction interval [Error bar]

Algorithm 1 is favored since it gets the point estimate almost identical to Algorithm 2.

Alg 1 for fast clinical use

Alg 2 for paper authors & rigorous research